

Program 3 : code a Dockerized Node js Applicatoin :

Step 1 :

Project structure and Docker file :

npm init -y ----- > to initialize a Node js application

npm init I ----- > to install an express nodejs frame work to develop restfull application

```
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab$ cd program3
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program3$ ls
Dockerfile  node_modules  package.json  package-lock.json  server.js
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program3$ cat Dockerfile
#BASE IMAGE
FROM node:alpine
```

```
#name of the maintainer , project and its version
LABEL maintainer="xxx@gmail.com"
LABEL Version="1.0"
LABEL projet="simple node app"
```

```
#speicifying the working directotry in the built image
WORKDIR /app
```

```
#copying all the package files for clean instaltion
COPY package*.json .
```

```
#npm clean install to install all the dependency
RUN npm ci
```

```
#copying the rest of the files to the working directory
COPY . .
```

```
#this is a simple docker file
```

```
#exposing port 3000
EXPOSE 3000
```

```
CMD ["node","server.js"]
```

Step 2 : Build the dockerfile to an image

command : docker build -t program3 .

```
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program3$ docker build -t program3 .
[+] Building 3.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 569B
=> [internal] load metadata for docker.io/library/node:alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:alpine@sha256:7e467cc5aa91c87e94f93c4608cf234ca24aac3ec941f7f3db207367ccccdd11
=> [internal] load build context
=> => transferring context: 44.36kB
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY package*.json .
=> CACHED [4/5] RUN npm ci
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:f763c549d3a795f7249f9b4b8a7dea4e548e974daf8fa93bd391e54e6ca53149
=> => naming to docker.io/library/program3
```

Step 3 : Run the docker container

Command : docker run -p 3000:3000 program3

-p for mapping the host port with the container port

program3 is the image of the nodejs application

```
abhi@abhi-ThinkPad-T490:~/Desktop/DevopsLab/program3$ docker run -p 3000:3000 program3
Server is Running in http://localhost:3000
```



Verifying it in the browser :

